Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

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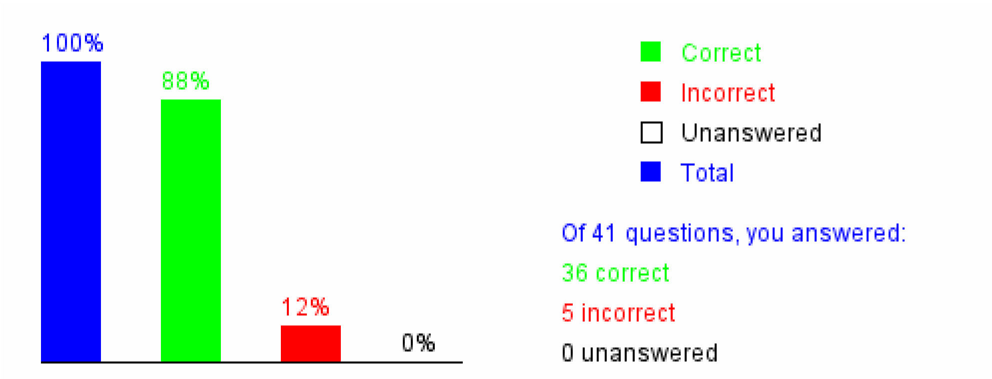
Chapter 2 Elementary Programming



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using. Thanks!

Section 2.3 Reading Input from the Console

2.1 What function do you use to read a string?

- (selected) A. `input("Enter a string")`
- B. `eval(input("Enter a string"))`
- C. `enter("Enter a string")`
- D. `eval(enter("Enter a string"))`

Your answer is correct

2.2 What is the result of eval("1 + 3 * 2")?

- A. `"1 + 3 * 2"`
- (selected) B. `7`
- C. `8`
- D. `"1 + 6"`

Your answer is correct

2.3 If you enter 1 2 3 in three separate lines, when you run this program, what will be displayed?

```
print("Enter three numbers: ")
number1 = eval(input())
number2 = eval(input())
number3 = eval(input())

# Compute average
average = (number1 + number2 + number3) / 3

# Display result
print(average)
```

- (selected) A. `1.0`
- B. `2.0`
- C. `3.0`

- D. 4.0

Your answer A is incorrect
Click here to show the correct answer

2.4 _______ is the code in natural language mixed with some program code.

- A. Python program
- B. A Python statement
- C. Pseudocode (selected)
- D. A flowchart diagram

Your answer is correct

2.5 If you enter 1 2 3 in one line, when you run this program, what will happen?

```
print("Enter three numbers: ")
number1 = eval(input())
number2 = eval(input())
number3 = eval(input())

# Compute average
average = (number1 + number2 + number3) / 3

# Display result
print(average)
```

- A. The program runs correctly and displays 1.0
- B. The program runs correctly and displays 2.0
- C. The program runs correctly and displays 3.0
- D. The program runs correctly and displays 4.0
- E. The program will have a runtime error on the input. (selected)

Your answer is correct

2.6 You can place the line continuation symbol __ at the end of a line to tell the interpreter that the statement is continued on the next line.

- A. /
- B. \
- C. # (selected)
- D. *
- E. &

Your answer C is incorrect
Click here to show the correct answer

Section 2.4 Identifiers

2.7 An identifier cannot be a keyword?

- A. true (selected)
- B. false

Your answer is correct

2.8 An identifier can contain digits, but cannot start with a digit?

- (●) A. `true`
- ( ) B. `false`

Your answer is correct

2.9 Which of the following is a valid identifier?

- [ ] A. `$343`
- [x] B. `mile`
- [ ] C. `9X`
- [ ] D. `8+9`
- [x] E. `max_radius`

Your answer is correct

2.10 Which of the following is a valid identifier?

- [ ] A. `import`
- [x] B. `mile1`
- [x] C. `MILE`
- [ ] D. `(red)`
- [ ] E. `"red"`

Your answer is correct

Section 2.5 Variables, Assignment Statements, and Expressions

2.11 If you enter 1, 2, 3, in one line, when you run this program, what will be displayed?

```
number1, number2, number3 = eval(input("Enter three numbers: "))

# Compute average
average = (number1 + number2 + number3) / 3

# Display result
print(average)
```

- ( ) A. `1.0`
- (●) B. `2.0`
- ( ) C. `3.0`
- ( ) D. `4.0`

Your answer is correct

2.12 What will be displayed by the following code?

```
x = 1
x = 2 * x + 1
print(x)
```

- ( ) A. `0`
- ( ) B. `1`
- ( ) C. `2`
- (●) D. `3`
- ( ) E. `4`

Your answer is correct

2.13 What will be displayed by the following code?

```
x = 1
x = x + 2.5
print(x)
```

A. `1`
B. `2`
C. `3`
D. `3.5`
E. `The statements are illegal`

Your answer is correct

Section 2.6 Simultaneous Assignments

2.14 What will be displayed by the following code?

```
x, y = 1, 2
x, y = y, x
print(x, y)
```

A. `1 1`
B. `2 2`
C. `1 2`
D. `2 1`

Your answer is correct

2.15 To following code reads two number. Which of the following is the correct input for the code?

```
x, y = eval(input("Enter two numbers: "))
```

A. `1 2`
B. `"1 2"`
C. `1, 2`
D. `1, 2,`

Your answer is correct

Section 2.8 Numeric Data Types and Operators

2.16 What is the result of 45 / 4?

A. `10`
B. `11`
C. `11.25`
D. `12`

Your answer is correct

2.17 In the expression 45 / 4, the values on the left and right of the / symbol are called ____.

A. `operators`
B. `operands`
C. `parameters`
D. `arguments`

Your answer is correct

2.18 What is the result of 45 // 4?

A. `10`
B. `11`
C. `11.25`
D. `12`

Your answer is correct

2.19 Which of the following expressions will yield 0.5?

A. `1 / 2`
B. `1.0 / 2`
C. `1 // 2`
D. `1.0 // 2`
E. `1 / 2.0`

Your answer is correct

2.20 Which of the following expression results in a value 1?

A. `2 % 1`
B. `15 % 4`
C. `25 % 5`
D. `37 % 6`

Your answer is correct
Click here to show an explanation

2.21 25 % 1 is _____

A. `1`
B. `2`
C. `3`
D. `4`
E. `0`

Your answer is correct

2.22 24 % 5 is _____

A. `1`
B. `2`
C. `3`
D. `4`
E. `0`

Your answer is correct

2.23 2 ** 3 evaluates to __________.

A. `9`

B. `8`
C. `9.0`
D. `8.0`

Your answer is correct

2.24 2 ** 3.0 evaluates to __________.

A. `9`
B. `8`
C. `9.0`
D. `8.0`

Your answer is correct

2.25 2 * 3 ** 2 evaluates to __________.

A. `36`
B. `18`
C. `12`
D. `81`

Your answer is correct

2.26 What is y displayed in the following code?

```
x = 1
y = x = x + 1
print("y is", y)
```

A. `y is 0.`
B. `y is 1 because x is assigned to y first.`
C. `y is 2 because x + 1 is assigned to x and then x is assigned to y.`
D. `The program has a compile error since x is redeclared in the statement int y = x = x + 1.`

Your answer is correct
Click here to show an explanation

2.27 Which of the following is equivalent to 0.025?

- [ ] A. `0.25E-1`
- [x] B. `2.5e-2`
- [x] C. `0.0025E1`
- [x] D. `0.00025E2`
- [x] E. `0.0025E+1`

Your answer BCDE is incorrect
Click here to show the correct answer

2.28 If a number is too large to be stored in memory, it ____________.

A. `causes overflow`
B. `causes underflow`
C. `causes no error`

D. `cannot happen in Python`

Your answer is correct

Section 2.9 Evaluating Expressions and Operator Precedence

2.29 What is the result of evaluating 2 + 2 ** 3 / 2?

A. `4`
B. `6`
C. `4.0`
D. `6.0`

Your answer is correct

Section 2.10 Augmented Assignment Operators

2.30 What is the value of i printed?

```
j = i = 1
i += j + j * 5
print("What is i?", i)
```

A. `0`
B. `1`
C. `5`
D. `6`
E. `7`

Your answer C is incorrect
Click here to show the correct answer

2.31 What is x after the following statements?

```
x = 1
x *= x + 1
```

A. `x is 1`
B. `x is 2`
C. `x is 3`
D. `x is 4`

Your answer is correct

2.32 What is x after the following statements?

```
x = 2
y = 1
x *= y + 1
```

A. `x is 1.`
B. `x is 2.`
C. `x is 3.`
D. `x is 4.`

Your answer C is incorrect
Click here to show the correct answer

2.33 To add a value 1 to variable x, you write

- [ ] A. `1 + x = x`
- [x] B. `x += 1`
- [ ] C. `x := 1`
- [x] D. `x = x + 1`
- [x] E. `x = 1 + x`

Your answer is correct

2.34 Which of the following statements are the same?

```
(A) x -= x + 4
(B) x = x + 4 - x
(C) x = x - (x + 4)
```

- ( ) A. `(A) and (B) are the same`
- (●) B. `(A) and (C) are the same`
- ( ) C. `(B) and (C) are the same`
- ( ) D. `(A), (B), and (C) are the same`

Your answer is correct

2.35 To add number to sum, you write (Note: Python is case-sensitive)

- [ ] A. `number += sum`
- [ ] B. `number = sum + number`
- [ ] C. `sum = Number + sum`
- [x] D. `sum += number`
- [x] E. `sum = sum + number`

Your answer is correct

2.36 Suppose x is 1. What is x after x += 2?

- ( ) A. `0`
- ( ) B. `1`
- ( ) C. `2`
- (●) D. `3`
- ( ) E. `4`

Your answer is correct

2.37 Suppose x is 1. What is x after x -= 1?

- (●) A. `0`
- ( ) B. `1`
- ( ) C. `2`
- ( ) D. `-1`
- ( ) E. `-2`

Your answer is correct

2.38 What is x after the following statements?

```
x = 1
```

```
y = 2
x *= y + 1
```

- ○ A. x is 1
- ○ B. x is 2
- ◉ C. x is 3
- ○ D. x is 4

Your answer is correct

Section 2.11 Type Conversions and Rounding

2.39 Which of the following functions return 4.

- ○ A. `int(3.4)`
- ○ B. `int(3.9)`
- ○ C. `round(3.4)`
- ◉ D. `round(3.9)`

Your answer is correct

2.40 Which of the following functions cause an error?

- ☐ A. `int("034")`
- ☑ B. `eval("034")`
- ☑ C. `int("3.4")`
- ☐ D. `eval("3.4")`

Your answer is correct

Section 2.12 Case Study: Displaying the Current Time

2.41 The time.time() returns ________________ .

- ○ A. the current time.
- ○ B. the current time in milliseconds.
- ○ C. the current time in milliseconds since midnight.
- ○ D. the current time in milliseconds since midnight, January 1, 1970.
- ◉ E. the current time in milliseconds since midnight, January 1, 1970 GMT (the Unix time).

Your answer is correct